

Dr. Harvindra Singh (he)

CONTACT INFORMATION	Department of Mathematics Indian Institute of Technology, Ropar-140001, Punjab, India Mobile:+91-9027279494 Email: singh.harvindrahs@gmail.com staff.harvindra.singh@iitrpr.ac.in																				
PERSONAL INFORMATION	Nationality: Indian Date of Birth: 05 Jun 1992 Gender: Male Marital Status: Single																				
CURRENT AFFILIATION	I am an ANRF National Post-Doctoral Fellow in the Department of Mathematics at Indian Institute of Technology Ropar , Ropar-140001, India.																				
PHD ADVISOR	Dr. L. K. Balyan Department of Natural Sciences PDPM IIITDM Jabalpur Jabalpur-482005, India Email: balyan@iiitdmj.ac.in																				
DOCTORAL DEGREE DETAILS	• Area: Applications of the spectral methods to the time-dependent PDEs • Thesis Title: Robust high-order numerical techniques for solving time-dependent non-linear smooth and non-smooth PDEs using spectral methods • Thesis Defense: Done • Degree Awarded: Yes																				
EDUCATION	<table><tr><th>Degree</th><th>Year</th><th>Institution/University</th><th>Division</th></tr><tr><td>Ph.D. (Applied Mathematics)</td><td>2025</td><td>PDPM IIITDM Jabalpur, MP, India</td><td>First</td></tr><tr><td>M.Sc. (Mathematics)</td><td>2017</td><td>IIT Ropar, Punjab, India</td><td>First</td></tr><tr><td>B.Sc. (Mathematics, Physics, Chemistry)</td><td>2014</td><td>Central University HNBGU, UK, India</td><td>First</td></tr><tr><td>Intermediate (Mathematics, Science, Social Science, Hindi, English,)</td><td>2011</td><td>NIC Dhanauri, UK, India</td><td>First</td></tr></table>	Degree	Year	Institution/University	Division	Ph.D. (Applied Mathematics)	2025	PDPM IIITDM Jabalpur, MP, India	First	M.Sc. (Mathematics)	2017	IIT Ropar, Punjab, India	First	B.Sc. (Mathematics, Physics, Chemistry)	2014	Central University HNBGU, UK, India	First	Intermediate (Mathematics, Science, Social Science, Hindi, English,)	2011	NIC Dhanauri, UK, India	First
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RESEARCH EXPERIENCE	• Currently working as an ANRF National Postdoctoral Fellow, conducting research in spectral methods, numerical analysis, and scientific computing, with a focus on the development and analysis of high-accuracy numerical techniques for differential equations. • I conducted research for my M.Sc. final-year project on “<i>Oscillation quenching in Allee effect induced metapopulation model</i>” under the supervision of Prof. Partha Sharathi Dutta at the Department of Mathematics, Indian Institute of Technology Ropar, Punjab, India.																				

PUBLICATIONS

1. **Harvindra Singh**, L. K. Balyan, *Fast Algorithm for Filter-Enhanced Polynomial Collocation Approach Addressing Discontinuous and Steep Gradient Solutions*, [International Journal of Computer Mathematics](#) (2026).
2. **Harvindra Singh**, and L. K. Balyan, *An efficient high order solver for multidimensional nonlinear time-evolution (Burgers) equation based on Legendre spectral method*, [Computational and Applied Mathematics](#) (2026).
3. **Harvindra Singh**, and L. K. Balyan, *Computationally Efficient Solver for the Quasi-Linear Burgers Equation Using the Legendre Pseudospectral Method*, [International Journal of Applied and Computational Mathematics](#) (2025).
4. **Harvindra Singh**, L. K. Balyan, A. K. Mittal, and P. Saini, *A numerically robust and stable time-space pseudospectral approach for multidimensional generalized Burgers-Fisher equation*, [Mathematics and Computers in Simulation](#) (2024).
5. **Harvindra Singh**, A. K. Mittal, and L. K. Balyan^{1*}, *Pseudospectral analysis for multidimensional fractional Burgers equation based on Caputo fractional derivative*, [Arabian Journal of Mathematics](#) (2024).
6. Mittal A. K., Balyan L. K., Panda M. K., Shrivastava P., and **Singh Harvindra**, *Pseudospectral analysis and approximation of two-dimensional fractional cable equation*, [Mathematical Methods in the Applied Sciences](#) (2021).

CONFERENCE & WORKSHOP

- **Harvindra Singh**, and L. K. Balyan, *An Efficient Pseudospectral Approximation of Non-Linear Time-Evolution Equations*, [Computational Techniques and Applications Conference CTAC2024](#) held at **Monash University, Australia**, during November 19-22, 2024.
- **Harvindra Singh**, and L. K. Balyan, *Improved Pseudospectral Approximation for Non-Linear Parabolic Equation*, [Nonlinear Analysis and Computational Techniques - \(ICNACT 2024\)](#), VIT Bhopal in collaboration with UTM Mexico, (2024). **(Accepted)**.
- **Harvindra Singh**, and L. K. Balyan, *An improved Chebyshev collocation approximation of the Burgers-Fisher equation*, [International Conference on Applied Mathematics and Mechanics \(ICAMM 2023\)](#), IIT Indore, (2023). **(Accepted)**.
- Presented a paper entitled "Spectral Approximation For Elliptic Eigenvalue Problem On Non-Smooth Domain" in the "[International Conference on Nonlinear Analysis and Applications-2022](#)", Organized by The Department of Mathematics, Assam Don Bosco University, India in collaboration with the Institute of Physics and Mathematics, Technological University of the Mixteca, Mexico and Department of Mathematics, Ramniranjan Jhunjhunwala College (Autonomous), Mumbai, India, held during 22-23 November 2022.
- Attended a workshop on "[Finite Volume and Spectral Methods for Hyperbolic Problems](#)", Centre for Applicable Mathematics, **TIFR, Bangalore**, from 4 to 15 Dec 2023.
- Participated in the Postgraduate Level Training Program at IIT Ropar on "[Differential Equations under the auspices of National Program on Differential Equations: Theory, Computation and Applications \(NPDE-TCA\)](#)" sponsored by DST, 2016.

AWARDS AND
FELLOWSHIPS

I have been honored to receive several awards and fellowships that have recognized my academic achievements and potential for research excellence. These include:

- Conferred with **Young Scientist Award** in the discipline of Mathematical Sciences during 37th **M.P. Young Scientist Congress** hosted by Mahatma Gandhi Chitrakoot Gramodaya Vishwavidyalaya Chitrakoot. The award included a cash prize of INR 25,000/-.
- Qualified **Graduate Aptitude Test in Engineering** (GATE 2018 & 2019), organized by Indian Institute of Technologies (IITs) & IISc on behalf of NCB-GATE for Department of Higher Education, MHRD.
- Qualified **Joint CSIR-UGC-NET** (2020 & 2021) conducted by the Council of Scientific & Industrial Research CSIR-UGC all-India Examination.
- Received **provisional offer under INSPIRE fellowship [DST/INSPIRE/03/2017/001914]** to pursue research in basic and applied sciences.
- Qualified **Joint Admission Test for M.Sc.** (JAM 2015), all-India level online entrance exam conducted by the Indian Institute of Technology, Guwahati.
- **INSPIRE Scholarship** during B.Sc. (2011-2014) and M.Sc. (2015-2017) studies, awarded by the Department of Science and Technology (DST) under the Ministry of Science and Technology, Indian Government.
*This scholarship is awarded to exceptional students who have obtained **top 1% aggregate marks in their Class XII examination** from any State/Central Education Board, to support their pursuit of higher education.*
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TEACHING
EXPERIENCE

- **Assistant Professor** in the department of mathematics **Dev Bhoomi Uttarakhand University, Dehradun, India** from **Oct 2024 - Nov 2025**.
- **Assistant Professor** in the department of mathematics (**Chandigarh University**) from **July 2018 – Aug 2019**, Punjab, India.
- Teaching Assistant for the following **undergraduate courses** during Ph.D.
Mathematics-I (Calculus) (**NS-1001**)
Mathematics-II (ODEs & PDEs) (**NS103a**)
Mathematics-II (Probability & Linear algebra) (**NS103b**)
Numerical Methods for Engineers (**NS-205f**)

COMPUTATIONAL
TECHNIQUES

MATLAB - apply for numerically solving the obtained system of ordinary differential equations and system of algebraic equations after pseudospectral discretization of the considered equations.

EXPERIENCE AND
SKILLS

- Teaching experience as an assistant professor in Dev Bhoomi Uttarakhand University (Oct 2024 - till date).
- Teaching experience as an assistant professor in Chandigarh University (July 2018 - Aug 2019).
- Teaching experience during Ph.D.
- Self-motivated programmer, experienced in MATLAB.

	• Good interest in research oriented projects. • Work experience on utilities like M.S. Power Point, M.S. Word, Excel, $\text{\LaTeX}$, MATLAB, Mathematica etc. • Miscellaneous: Experience in Linux and Windows operating systems.
POSITION OF RESPONSIBILITIES	• Active member in <i>IIC (Institute Innovation Council)</i>, IIITDM Jabalpur, India, from 2020-2022. • Mess committe convener during Dec 2019 - Jun 2022 • Hosted event as an anchor in “National Conference on Recent Trends in Physical Sciences (NCRTPS-2024)” organized by Natural Science Discipline (Mathematics), PDPM IIITDM Jabalpur, India on the occasion of “National Science Day 2024” during 28 Feb, 2024.
EXTRACURRICULAR ACTIVITIES	• Won first prize in “Matrabhumi kavita path and geet gayan” held at IIT Ropar during Post doc (2026). • Won first prize in “Deshbhakti kavita path and geet gayan” held at IIT Ropar during Post doc (2026). • Won first prize in “PATRIOTISM SINGING COMPITITION” held in IIT Ropar during MSc (2016). • Secured second prize in the Cultural Festival of IIITDM Jabalpur (TARANG) in 2021 for a singing band performance. • Participated in “SOLO SINGING” IIT inter college fest “ZIETGIEST” of IIT, Ropar (2017). • Won prize in an essay writing competition on the topic ‘Contributions of Indians towards Freedom Struggle’ during Ph.D. in March 2021. • Member of M.Sc. & Ph.D. cricket team. • Scored a century in the FIT India Cricket Tournament 2022 - <i>The first ever by a Ph.D. scholar since the tournament's inception.</i>
REFERENCES	1. Dr. L. K. Balyan Faculty of Mathematics & head of the department Department of Natural Science PDPM IIITDM Jabalpur Jabalpur - 482005, India Phone No.: +91-761-2794346 Email: balyan@iiitdmj.ac.in lokendra.balyan@gmail.com 2. Dr. Manoranjan Mishra Professor Department of Mathematics Indian Institute of Technology Ropar Rupnagar, Punjab - 140001, India Phone No.: 24-2158 Email: manoranjan@iitrpr.ac.in

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3. [Dr. Partha Sharathi Dutta](#)
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